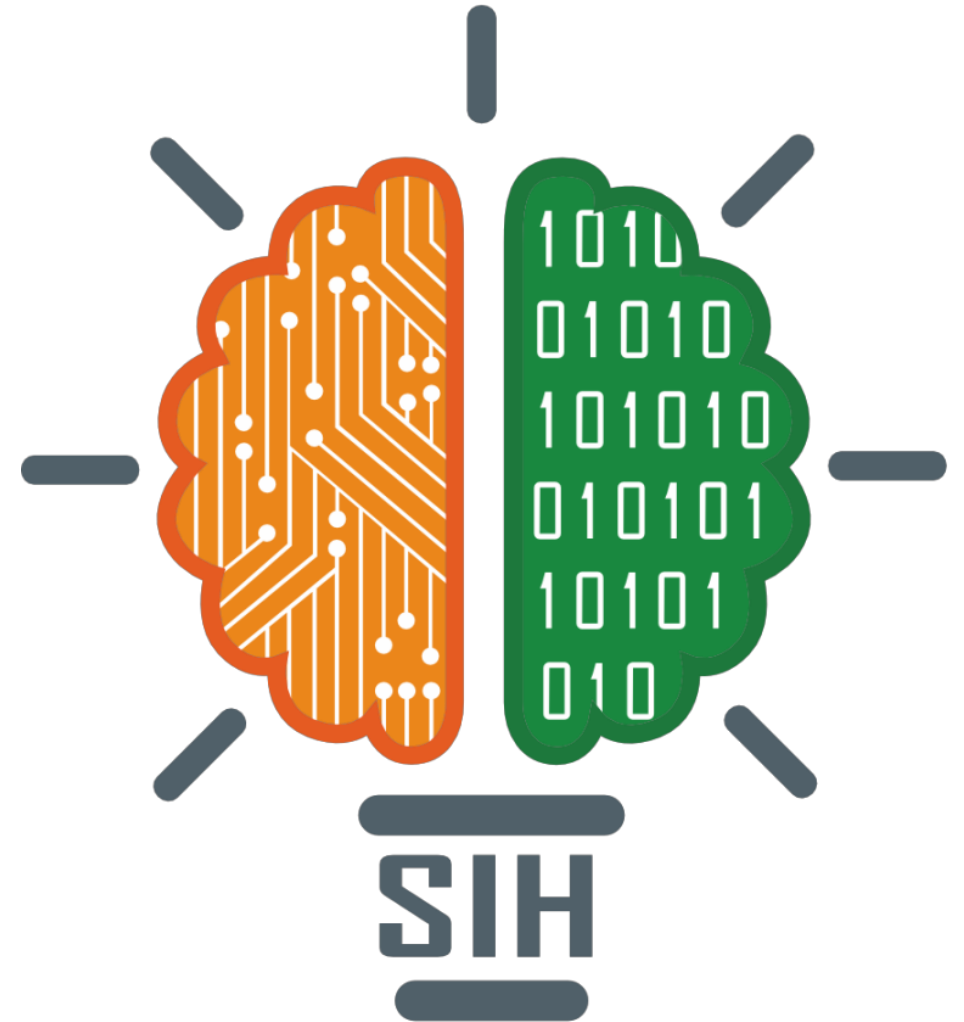


Adaptive Foveated 2.5D Semantic Mapping for Autonomous Navigation

- **Problem Statement ID** – 26053
- **Problem Statement Title** – Adaptive Foveated 2.5D Semantic Mapping from LiDAR Point Clouds for Autonomous Navigation
- **Theme** – Smart Vehicles
- **PS Category** – Software
- **Team ID** – 137735
- **Team Name** – TeraPulse
- **Team Members** – Abhay Verma (Team Leader), Lokendra Singh, Yashvardhan Jain, Chetan Meena, Sachin Chaubey, Esha Verma
- **Institute** – Indian Institute of Information Technology Bhopal



Adaptive Foveated 2.5D Semantic Mapping for Autonomous Navigation

❖ Proposed Solution

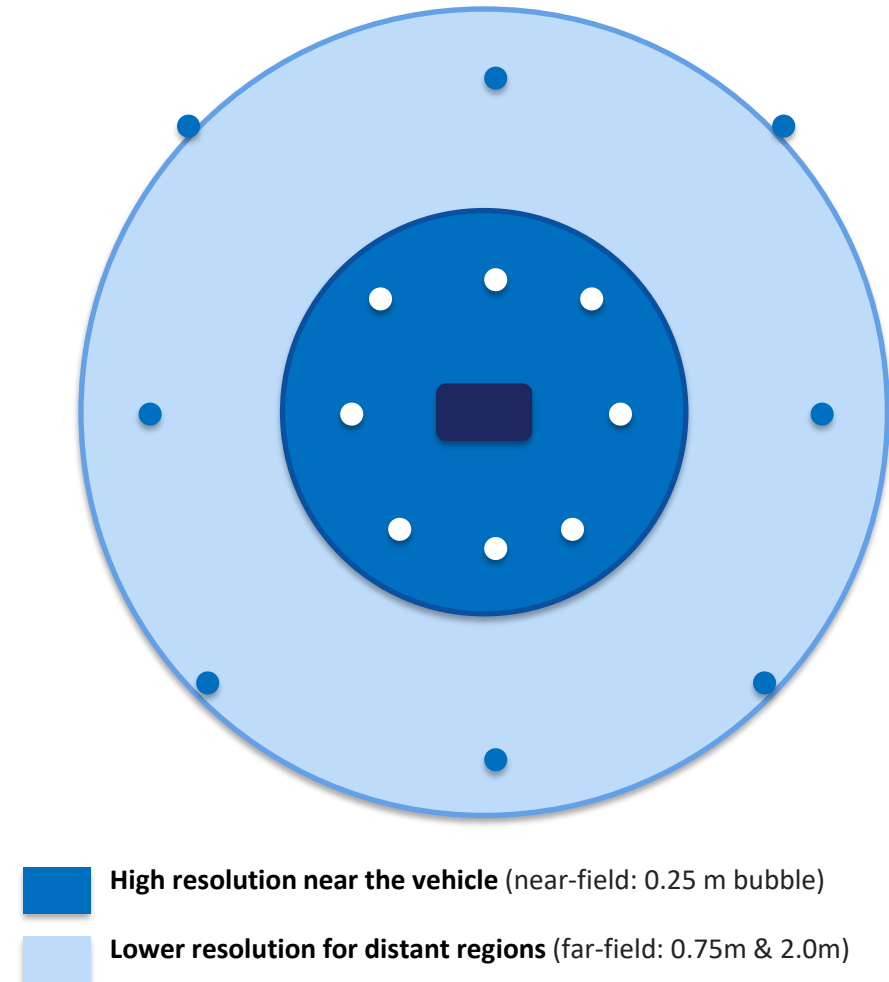
- **2.5D Semantic Elevation Map:** Generated directly from raw LiDAR point clouds.
- **High-Precision Near Field:** Fine resolution (0.25 m) near vehicle for critical steering.
- **Bandwidth-Saving Far Field:** Lower resolution (0.75 m & 2.0 m) for distant regions — cuts unnecessary compute.
- **Elevation & Slope Awareness:** Preserves terrain slopes, curbs & ditches without full 3D voxels.
- **Real-Time Terrain Intelligence:** Identifies drivable ground, static obstacles & dynamic objects.
- **Autonomous Navigation Ready:** Enables instant A* path planning & Pure Pursuit motion control.

❖ Innovation & Uniqueness

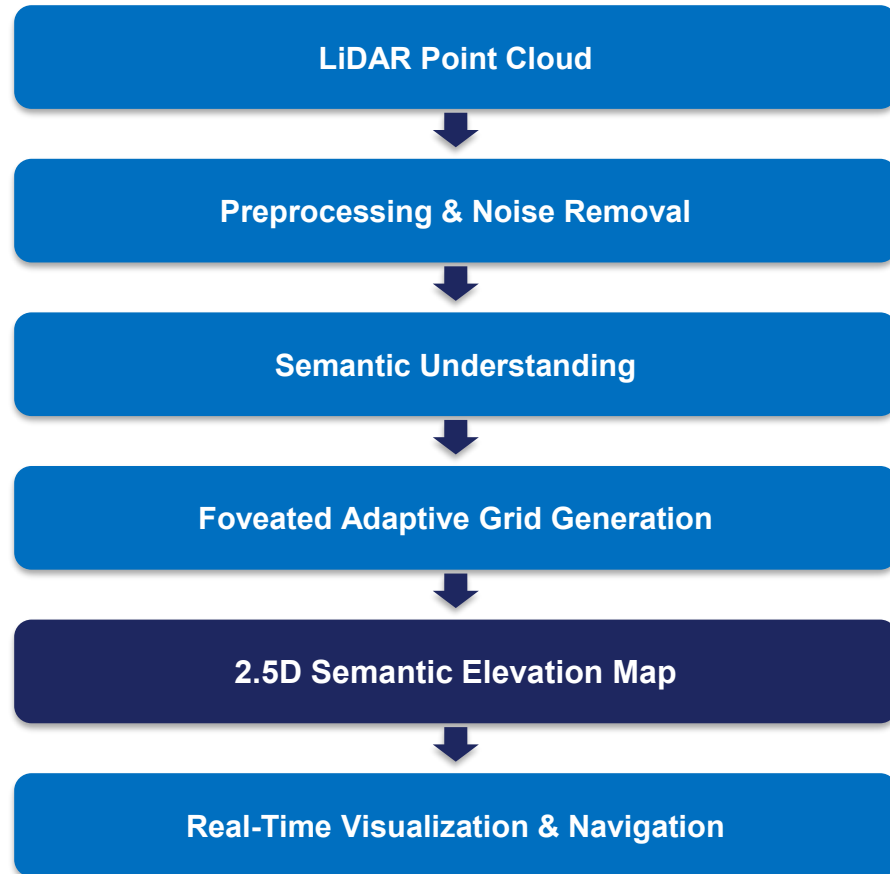
Foveated adaptive resolution *focuses computational resources where they matter most, instead of processing the entire environment at the same resolution.*

- **Measurable Impact:** Slashes cell memory by 66.2% and cuts processing latency to 24.23 ms (>35 FPS).

Foveated 2.5D Semantic Elevation Map

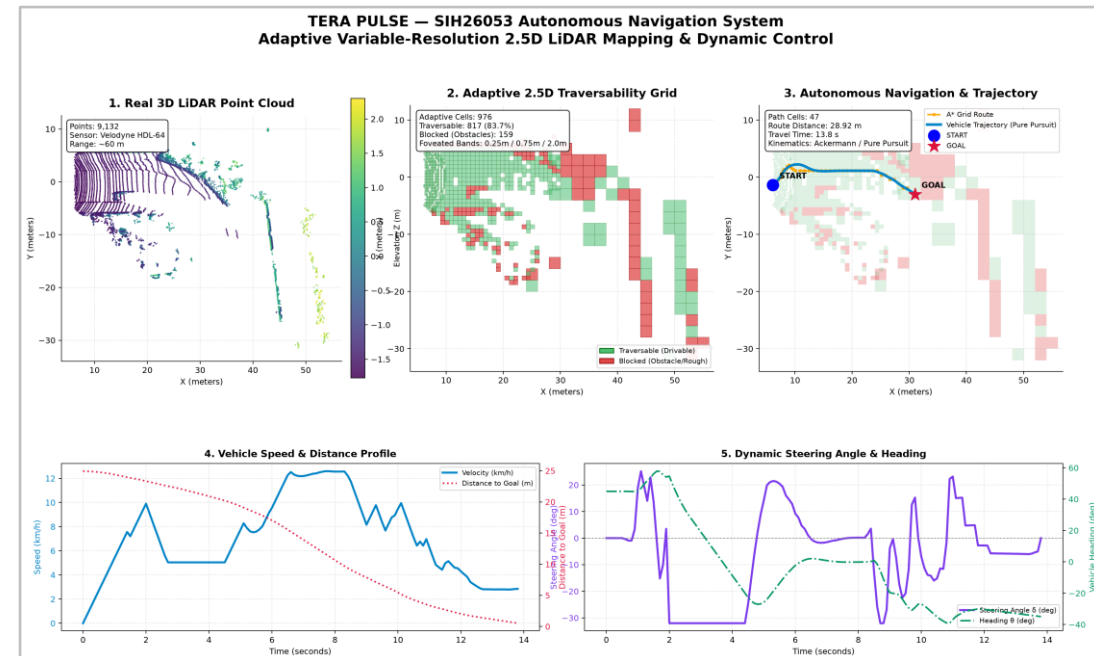


System Workflow



Technologies & Implementation

- **Python, Open3D & NumPy:** Point cloud preprocessing, voxelization & spatial hashing.
- **Deep Learning:** Semantic terrain classification (Ground, Static Obstacles, Dynamic Vehicles).
- **A* Search & Pure Pursuit:** Collision-free path planning & smooth 10 Hz vehicle steering kinematics.
- **Interactive Web Dashboard:** Flask REST API, 60 FPS HTML5 Canvas with moving foveation bubble.



Feasibility

- **Standard Datasets:** Uses widely available LiDAR point cloud datasets & open-source tools.
- **Rapid Development:** Python, Open3D, and PyTorch enable agile modular prototyping.
- **Low-Power Efficiency:** Adaptive resolution drastically reduces memory & compute requirements.
- **Modular Pipeline:** Components tested independently with 5/5 automated unit tests passing.

Challenges

- **Heavy Data Stream:** Real-time processing of large LiDAR point clouds at 10–20 Hz.
- **Off-Road Accuracy:** Accurate semantic classification in unstructured off-road scenes.
- **Underpasses & Tunnels:** Avoiding false-positive obstacle detection for bridge ceilings.
- **Detail Retention:** Maintaining critical safety clearance while reducing distant resolution.

Mitigation Strategy

- **Foveated Attention:** Variable-resolution mapping concentrates compute in near-field (24.23 ms).
- **SOR & RANSAC Fit:** Statistical Outlier Removal & ground fit eliminate dust & sensor reflections.
- **Vertical Clearance:** Clearance filter ignores points >2.2 m, enabling safe tunnel traversal.
- **Proven Benchmark:** Benchmarked on 20,672 points at >35 FPS with zero crash failures.



Impact

- **Tactical UGV Navigation:** Supports safer and faster autonomous military navigation.
- **Critical Near-Field Safety:** Sub-meter precision in safety-critical perimeter around vehicle.
- **Zero-Visibility Ops:** Active LiDAR penetrates smoke, dust, fog & battlefield darkness.
- **Terrain Intelligence:** Real-time elevation slope, ditches & surface roughness understanding.

Applications

- **Defence & Tactical UGVs:** Logistics convoys, perimeter patrols, scout rovers & minefield recon.
- **Off-Road Autonomous Rovers:** Search-and-rescue rovers in disaster zones, agricultural robotics.
- **Driver Assistance (ADAS):** Intelligent terrain HUD and active collision prevention in dense fog.

Benefits

- **66.2% Memory Reduction:** Stored active cells drop from 2,888 to 976 via foveated concentration.
- **4.8× Faster Latency:** Down from 115.94 ms to 24.23 ms — exceeds 35 FPS real-time vehicle requirement.
- **Full 2.5D Elevation Profile:** Preserves critical ground slope, curbs & height without voxel clutter.
- **Verified Path Traversability:** 817 safe corridor cells verified with smooth 10 Hz Pure Pursuit steering.

Evaluation Metric	Uniform Grid	TeraPulse Foveated	Practical Advantage
Active Stored Cells	2,888 cells	976 cells	66.2% RAM Saved
Compute Latency	115.94 ms (8.6 FPS)	24.23 ms (>35 FPS)	4.8× Faster Control
Hardware Footprint	High-End Desktop GPU	Embedded CPU / Jetson	Low Power UGV Edge
Path Generation	Delayed Waypoints	10 Hz Dynamic Telemetry	Safe Agile Steering

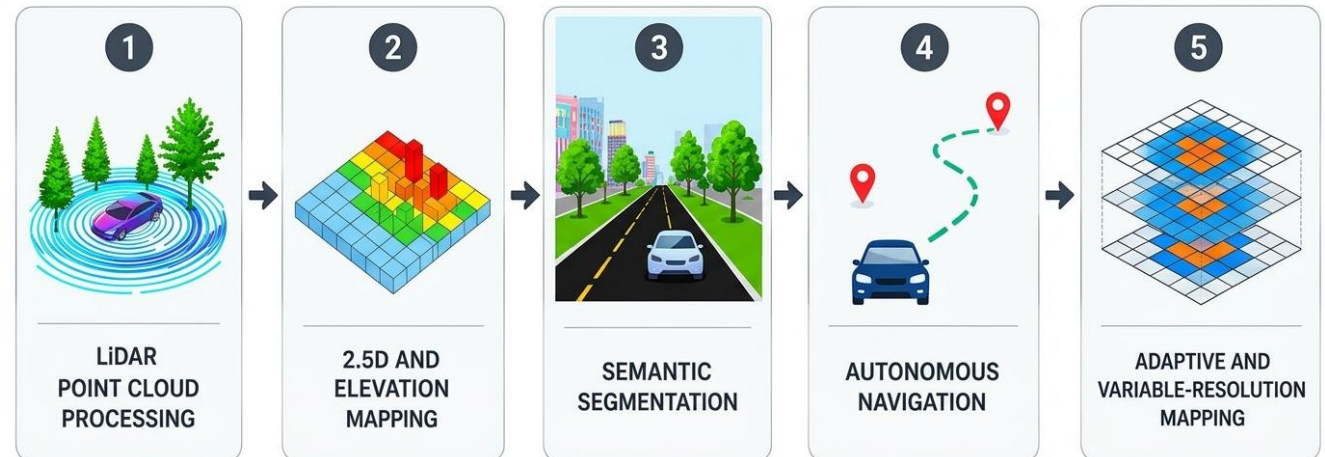
Research Areas

- **LiDAR Point Cloud Processing:** Statistical outlier removal & RANSAC ground plane fit.
- **2.5D Elevation Mapping:** Robot-centric height & slope mapping for rough terrain.
- **Semantic Segmentation:** Deep learning segmentation of drivable vs obstacle cells.
- **Autonomous Navigation:** A* heuristic planning & Pure Pursuit vehicle kinematics.
- **Adaptive Spatial Hashing:** Foveated variable-resolution concentric indexing.

References

- **KITTI Vision Suite:** Geiger et al., Velodyne HDL-64E LiDAR benchmark.
- **ETH Zurich Mapping:** P. Fankhauser et al. (IEEE IROS), Robot-Centric Elevation Mapping.
- **CMU Pure Pursuit:** R. C. Coulter (Carnegie Mellon), Pure Pursuit Path Tracking.
- **Point Cloud ML:** Q. Hu et al. (CVPR), RandLA-Net Point Cloud Segmentation.
- **Open-Source Tools:** Open3D, PyTorch, NumPy, Flask REST APIs.

LiDAR 2.5 RESEARCH PIPELINE



Project Deliverables & Repository

- **Interactive Web Simulator:** Flask REST API + 60 FPS HTML5 Canvas with live vehicle foveation.
- **Automated Test Suite:** 5/5 unit tests (tests/test_pipeline.py) passing in 4.0s.
- **GitHub Repository:** github.com/AbhayVerma628/SIH26053_Lidar (Open Source MIT License).
- **Telemetry & Artifacts:** 5-panel dashboard, 10 Hz vehicle telemetry CSV & planned A* waypoints.